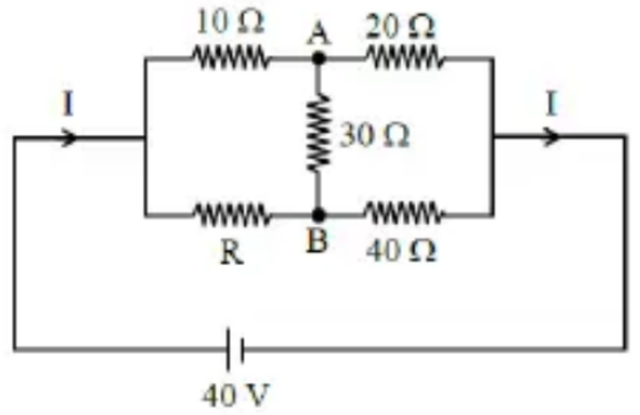


As $x = pt + qt^2 + rt^3$,

$$v = \frac{dx}{dt} = p + 2qt + 3rt^2 \text{ and } a = \frac{dv}{dt} = 2q + 6rt = 8 + 30t$$

Initial acceleration at $t = 0$ is $a_i = 8\ m/s^2$

$F_i = (2\ kg)(8\ m/s)^2 = 16\ N$

Power generated $= \frac{90}{100} \times \frac{dm}{dt} \times gh = \frac{90}{100} \times 15 \times 10 \times 60 = 8100\,w$

$$mg = 400 \Rightarrow m(g + a) = 600$$

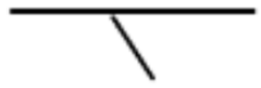
$$(g + a) = \frac{600}{400} g \Rightarrow a = 0.5g = 4.9 ms^{-2}$$

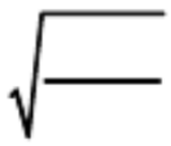
$$E_X = \frac{-\partial V}{\partial X} = -4x \qquad E_Y = -\frac{\partial V}{\partial Y} = +3$$

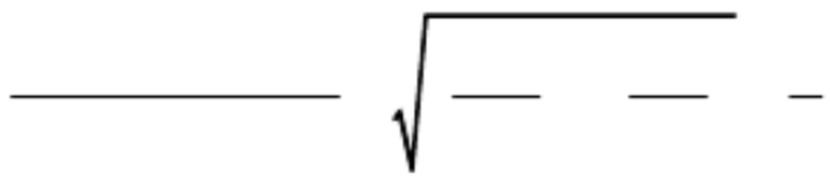
$$\overline{E} = E_X\hat{i} + E_Y\hat{j} = -4x\hat{i} + 3\hat{j} = 3\hat{j}\ N/C$$

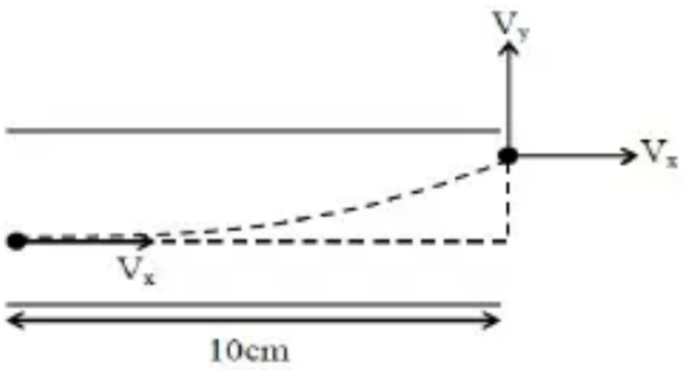
$$\frac{\rho_1}{\rho_2} = \frac{(1+\alpha t_1)}{(1+\alpha t_2)} \Rightarrow \frac{1}{2} = \frac{(1+0.00125\times 27)}{(1+0.00125\times t)} \Rightarrow t = 854°C \Rightarrow T = 1127\ K$$



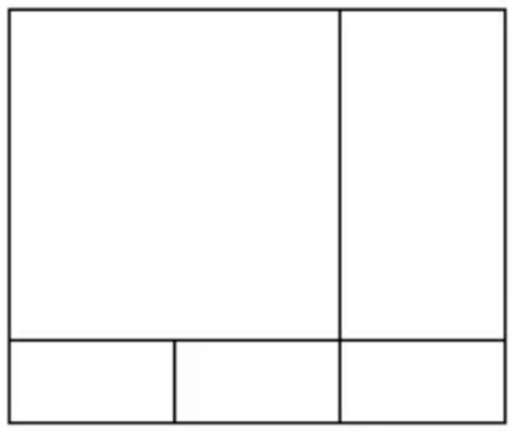


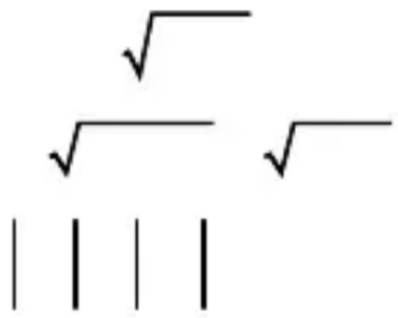





V_y
V_x
V_x
10cm

$\vec{\omega}$
$\vec{B}$





( )

( )

( )

$a_t = r\alpha = 8$; $a_c = r\omega^2 = 6$

( ) ( )

[ ]

[ ]

S $10\,W$ per $1L \Rightarrow \frac{10\,W}{M}$ moles/L

K_{sp} 10^5 108 $\frac{W^5}{M^5}$

63. Zn^{2}, Mn^{2}

64. Correct the reactions given in question as

$X_2 \rightleftharpoons X$ $X(\text{fast})$

X Y_2 XY $Y(\text{slow})$

Slow step is the rate determining step

Rate $k[X][Y_2]$

Equilibrium constant for fast step, K $\frac{[X]^2}{[x_2]}$

$[X]$ $\sqrt{K[X_2]}$

By substituting $[X]$ in equation (i), we get

Rate $k\sqrt{K[X_2]}[Y_2]$ $K[X_2]^{1/2}[Y_2]$

Order of reaction = $\frac{1}{2}$ 1 $\frac{3}{2}$ 1.5

65. Conceptual

66. $Fe(OH)_3$ is lyophobic colloid

67. D ( ) Fructofuranose

68. Acidic buffer is HCN + KCN

weak acid salt of its strong base

P^H P^{K_a} $\log \frac{N_s V_s}{N_a V_a}$

69. Aniline can not be prepared by Gabriel Phtalimide reaction

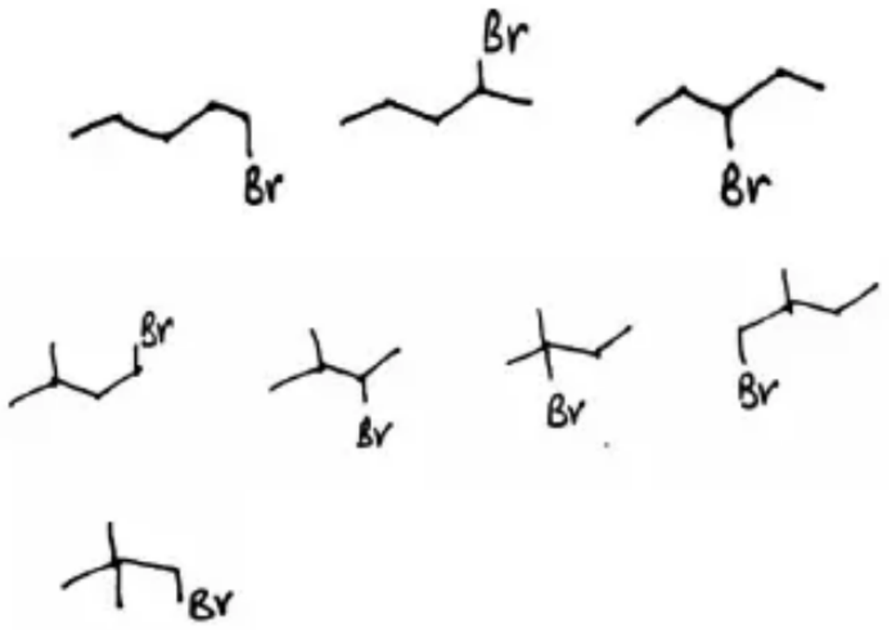
70. Group – 15 – Mc

Group – 16 – Lv

Group – 17 – Ts

Group – 15 – Og

71.



72. Sandmeyer's reaction proceeds by Nucleophilic substitution and can also proceed by free radical substitution

73. Conceptual

74.

CH_3 CH CH CH_3 $\overset{alkKMnO_4}{\underset{273K}{}}$

$CH_3-\underset{OH}{CH}-\underset{OH}{CH}-CH_3$

75. Conceptual

76. Conceptual

77. Conceptual

78. A = CH_3 CH_2 OH

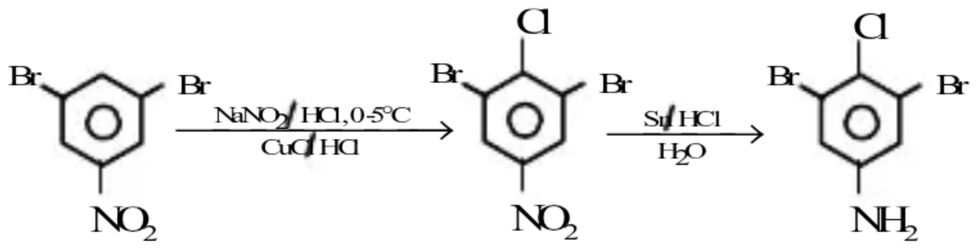
B = CH_3 CH_2 Cl

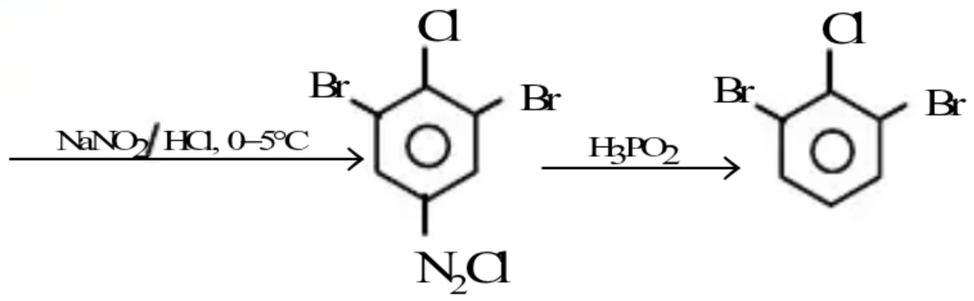
C = CH_2=CH_2, CH_3 CH_2 OH

Functional isomer of

CH_3 CH_2 OH is CH_3 O CH_3

79.





80. z = 32 is Ge - belong to 4th period, 14th group & is metalloid.

81. Conceptual

82. Conceptual

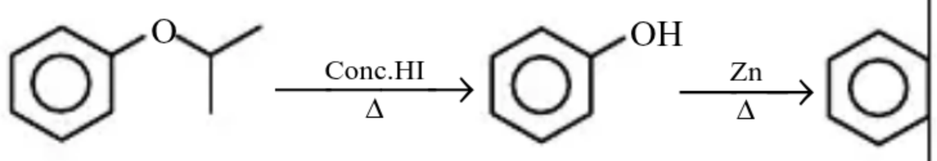
83. In anhydrous Ti^{3} splitting of d – orbitals does not occur due to absence of ligand. Hence there is no d – d transition

84. B) Ce $[Xe]\,6s^2 5d^1 4f^1$

C) Eu^3 Ce^3

85. Correct BP order is butan – 1 – o1 > butan – 2 – o1 > propan – 1 – o1

86.



87. A) P – amino azobenzene is yellow dye

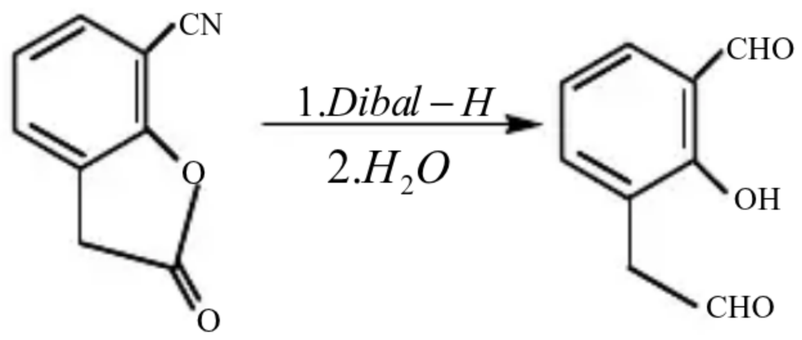
B)Yield in Sandmeyer > Gatterman reaction

C)Soluble due to acidic hydrogen

D) True

88. Conceptual

89.



Dibal –H can reduce both –CN and ester groups

90. Except O all others are -I groups

92. NCERT-Class-XII : 33, 34, 35.

The ovulatory phase is followed by the luteal phase

94. NCERT-Class-XII : 42.

96. NCERT-Class-XII : 207.

98. NCERT-Class-XI : 187.

100. NCERT-Class-XII : 47.

102. NCERT-Class-XII : 218.

103. NCERT-Class-XII : 240.

Posterior pituitary is under the direct neural regulation of the hypothalamus.

105. NCERT-Class-XII : 132.

108. NCERT-Class-XI : 199.

109. NCERT-Class-XI : 47.

Gnathus = Jaws, Stomata = Mouth

Animals whose mouth is supported by jaws are placed under Gnathostomata.

110. Old NCERT-Class-XI : 103.

113. NCERT-Class-XII : 76.

115. NCERT-Class-XI : 82.

117. NCERT-Class-XII : 141.

119. NCERT-Class-XII : 27, 30.

121. NCERT-Class-XI : 37, 39, 40.

Poriferans exhibit cellular level where as cnidarians exhibit tissue level of organization.

123. NCERT-Class-XII : 134.

124. NCERT-Class-XII : 231, 234.

126. NCERT-Class-XII : 117, 118.

129. NCERT-Class-XII : 44, 45.

130. NCERT-Class-XII : 223.

132. NCERT-Class-XII : 122.

134. NCERT-Class-XII : 183.

136. NCERT-Class-XII : 37.

139. NCERT-Class-XI : 212.

140. NCERT-Class-XII : 184.

143. NCERT-Class-XII : 104.

The functions are unknown for nearly 50 per cent of the discovered genes.

144. NCERT-Class-XII : 210, 211.

146. NCERT-Class-XII : 219.

Columbia located near the equator has nearly 1400 species of birds.

148. NCERT-Class-XII : 124.

150. NCERT-Class-XII : 225.

152. NCERT-Class-XII : 198, 199.

154. NCERT-Class-XII : 209.

156. NCERT-Class-XII : 74.

159. Old NCERT-Class-XI : 114.

160. NCERT-Class-XI : 207.

162. Old NCERT-Class-XI : 113.

164. NCERT-Class-XII : 195.

166. NCERT-Class-XI : 245.

168. NCERT-Class-XI : 50.

Neophron belongs to aves, it is oviparous and its development is direct.

170. NCERT-Class-XI : 189.

171. NCERT-Class-XI : 201.

The inner lining of squamous endothelium of arteries is called tunica intima.

173. NCERT-Class-XI : 246.

175. NCERT-Class-XI : 267.

177. NCERT-Class-XI : 44